

RITESH KUMAR

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE (AI/ML) student at Bharati Vidyapeeth College of Engineering (GGSIPU) with hands-on experience in end-to-end machine learning, NLP, deep learning, and full-stack AI development. Proven track record with a production ML internship at GAIL India, hackathon wins (1st place, ISA Delhi), and leadership at Google Developer Students Club. Proficient in Python, TensorFlow, Scikit-learn, and Explainable AI. Seeking Software Engineering, ML, Data Science, and AI internship opportunities to deliver data-driven, impactful solutions.

EDUCATION

Bharati Vidyapeeth College of Engineering (GGSIPU) — New Delhi, India 2023 – 2027
B.Tech in Computer Science Engineering (AI/ML Specialization)

Angels Public School — New Delhi, India 2023
CBSE Class XII

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

ML & AI Frameworks: TensorFlow, Scikit-learn, Keras, ResNet, Random Forest, SMOTE, NLP, TF-IDF, Cosine Similarity

Explainable AI (XAI): Grad-CAM, LIME

Data Science: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Hyperparameter Tuning

Web & Backend: FastAPI, Streamlit, REST APIs, Flask

Tools & Cloud: Git, GitHub, Docker, Google Cloud Platform (GCP), Linux

PROFESSIONAL EXPERIENCE

GAIL India Limited | New Delhi, India [\[GitHub\]](#) Jul 2025 – Aug 2025
Machine Learning Intern

- **Engineered** a production-ready House Price Prediction system using Linear, Ridge, and Lasso Regression, achieving $R^2 > 0.73$ and reducing RMSE by 10% via systematic hyperparameter tuning and EDA.
- **Built** robust data preprocessing pipelines incorporating missing value imputation, feature scaling, and outlier detection on the Boston Housing dataset, improving overall model stability.
- **Deployed** the end-to-end ML workflow as a real-time Streamlit web application, enabling interactive predictions for end-users with zero infrastructure overhead.

TECHNICAL PROJECTS

ADHD Detection System [\[GitHub\]](#) 2026

- **Designed** a state-of-the-art AI diagnostic system leveraging deep learning (ResNet architectures) to analyze multimodal neural signals (EEG + fMRI imagery) for automated ADHD screening.
- **Integrated** Explainable AI (XAI) frameworks — Grad-CAM and LIME — to generate transparent, clinically interpretable visualizations, reducing black-box AI limitations and increasing clinical trust.
- **Developed** an accessible web-based interface to streamline clinical workflows, minimize subjective observer bias, and deliver objective, automated diagnostic insights to clinicians.

Movie Recommendation System [\[GitHub\]](#) 2025

- **Architected** a content-based NLP recommendation engine using TF-IDF vectorization and Cosine Similarity to process metadata (genres, cast, crew, overviews) and return the top 5 personalized film recommendations.
- **Built** an interactive Streamlit web interface with real-time TMDB API integration to fetch and display movie posters, enhancing UX without relying on collaborative filtering or user history.

Customer Churn Prediction System [\[GitHub\]](#) 2024

- **Engineered** a classification-based ML framework using Random Forest algorithms to predict customer attrition with 85%+ accuracy, identifying key retention metrics for business stakeholders.
- **Mitigated** severe class imbalance using SMOTE (Synthetic Minority Over-sampling Technique) within robust data preprocessing pipelines, significantly improving minority-class recall.
- **Deployed** a full-stack web application with a FastAPI backend and dynamic Chart.js visualization dashboards, delivering real-time churn risk insights to decision-makers.

CERTIFICATIONS & TRAINING

- **Machine Learning Skill Up & Data Science Bootcamp** — GeeksforGeeks
- **AI Fluency for Students** — Anthropic

- **Python Programming** — GeeksforGeeks
- **Generative AI Architecture & Applications** — Ducat Training School (Industrial Training)
- **Enterprise Network Implementation** — Network Bulls (Industrial Workshop)

LEADERSHIP & ACHIEVEMENTS

- **Administration Executive — Google Developer Students Club (BVCOE)** | Sep 2025 – Apr 2026
- Coordinated technical workshops, seminars, and developer events, actively engaging a community of 200+ students and managing cross-functional team operations across multiple technical verticals.
- **Core Team Member — Venuva BVCOE (Event Management Society)** | Sep 2024 – Apr 2025
- Led end-to-end planning and execution of college-wide technical and cultural events, driving high student engagement and seamless cross-departmental collaboration.
- **Hackathon Achievements:** 1st Place — Innovate and Automate Hackathon (ISA Delhi Chapter, Hardware & Automation); 4th Place & Special Mention — Agentic AI Hackathon (IEEE MAIT) for innovative multi-agent AI system design; Runner-Up in 3 additional collegiate hackathons.
- **Competitive Engagement:** Active participant in 10+ hackathons and ideathons spanning AI, software engineering, and hardware domains; consistently recognized for creative problem-solving and technical depth.